Mehrshad Bagherebadian

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Education

Michigan State University East Lansing, MI
B.S. in Computer Science, Concentration in Artificial Intelligence *Expected Dec. 2026*
Minors: Data Science and CMSE; Business Cognate

Coursework: *AI/ML:* CSE 404 Machine Learning, CSE 440 Artificial Intelligence, CSE 446 AI Agents
Data: CSE 482 Big Data Analysis, CMSE 492 Applied ML, CMSE 381 Data Science Methods
Core: STT 442 Probability & Statistics II, CSE 435 Software Engineering, CSE 477 Web Development

Experience

Software Developer — The Capstone Experience Aug. 2026 – Present
Michigan State University — Urban Science *East Lansing, MI*

- Developing an AI-powered dealership analytics platform with Angular, TypeScript, C#/.NET, SQL Server, and Python/FastAPI; applying machine learning and AI agents to relate offer characteristics, customer distance, and service history to response and redemption rates, informing campaign targeting, pricing, and timing
- Building a multi-coupon extraction pipeline with Azure Document Intelligence OCR, Microsoft AI Foundry, LangChain, and LangGraph, combining structured LLM extraction, field-level validation, and human review
- Developing interactive dashboards and geographic market views to benchmark campaign effectiveness against competitors and in-network dealerships, with Docker deployment to an Azure VM through GitLab CI/CD

Undergraduate Researcher Aug. 2026 – Present
Michigan State University — Advisor: Dr. Mohammad M. Ghassemi *East Lansing, MI*

- Developing a lightweight NLP agent for confidence-aware biomedical evidence triage, combining evidence retrieval and question answering with selective answering to defer uncertain responses for human review
- Designing controlled experiments with complete, reduced, and noisy evidence to evaluate answer coverage versus reliability, including unsupported and overconfident responses
- Preparing *Evaluating Selective Answering in a Lightweight NLP Agent for Confidence-Aware Biomedical Evidence Triage* for submission to the EACL 2027 Student Research Workshop and presentation at MSU's UURAF

Projects

Claim/Case Timeline Builder | MSU AI Club | Auto-Owners Insurance Aug. 2026 – Present

- Developing a Python/FastAPI backend to convert adjuster notes, reports, correspondence, and payment records into chronological claim timelines for a Streamlit interface, linking every event to source evidence
- Building a RAG pipeline using Azure Document Intelligence OCR, LlamaIndex, ChromaDB, embeddings, and semantic retrieval for event extraction, date resolution, and deduplication into structured JSON/JSONL
- Collaborating with the testing team to evaluate event completeness, date accuracy, and chronological ordering against human-written timelines on synthetic claims

LogFlareX | *FastAPI, React, PostgreSQL, Docker, AWS* May. 2025 – Aug. 2026

- Built a log monitoring platform with FastAPI/PostgreSQL ingestion and filtering, configurable error alerts, and a React/TypeScript dashboard styled with Tailwind CSS; deployed with Docker on AWS and integrated Prometheus/Grafana for live metrics and system monitoring

InsightDesk | *Python, FastAPI, scikit-learn, React, TypeScript* Dec. 2024 - Mar. 2025

- Built an internal knowledge search tool using local TF-IDF retrieval to answer questions over notes and CSVs, returning source snippets and flagging low-confidence responses without a paid LLM API

Technical Skills

Languages: Python, C#, C++, TypeScript, JavaScript, SQL, R, HTML/CSS
Frameworks: Angular, React, Next.js, ASP.NET Core, FastAPI, Streamlit, SQLAlchemy, Tailwind CSS
AI & Data: LangChain, LangGraph, LlamaIndex, TensorFlow, PyTorch, scikit-learn, pandas, NumPy, Matplotlib, TF-IDF, RAG, OCR
Databases: SQL Server, PostgreSQL, ChromaDB, MySQL
Cloud & Tools: Azure, AWS, Microsoft AI Foundry, GCP, Docker, Git, GitLab CI/CD

Activities & Languages

Activities: Iranian Student Association, MSU AI Club, Imagine Software at MSU, Ping Pong Club
Languages: English, Persian/Farsi, Spanish (beginner)